CS23532-COMPUTER NETWORKS LAB MANUAL

EXPERIMENT-5 Develop a customized ping command to test the server connectivity

Aim - To develop a customized Ping program in Python that tests server connectivity by sending ICMP echo requests and displaying the response time and status.

Requirements

Hardware

• Laptop/Desktop

• Internet or Local Area Network (LAN)

Software

• Python 3.8 or above

• Visual Studio Code (VS Code) / IDLE / PyCharm

• Windows 10/11 or Ubuntu Linux.

custom_ping.py - C:/Users/MAHATHI A J/Documents/CustomPing/custom_ping.py (3.14.0)

File Edit Format Run Options Window Help

```
import subprocess
import platform

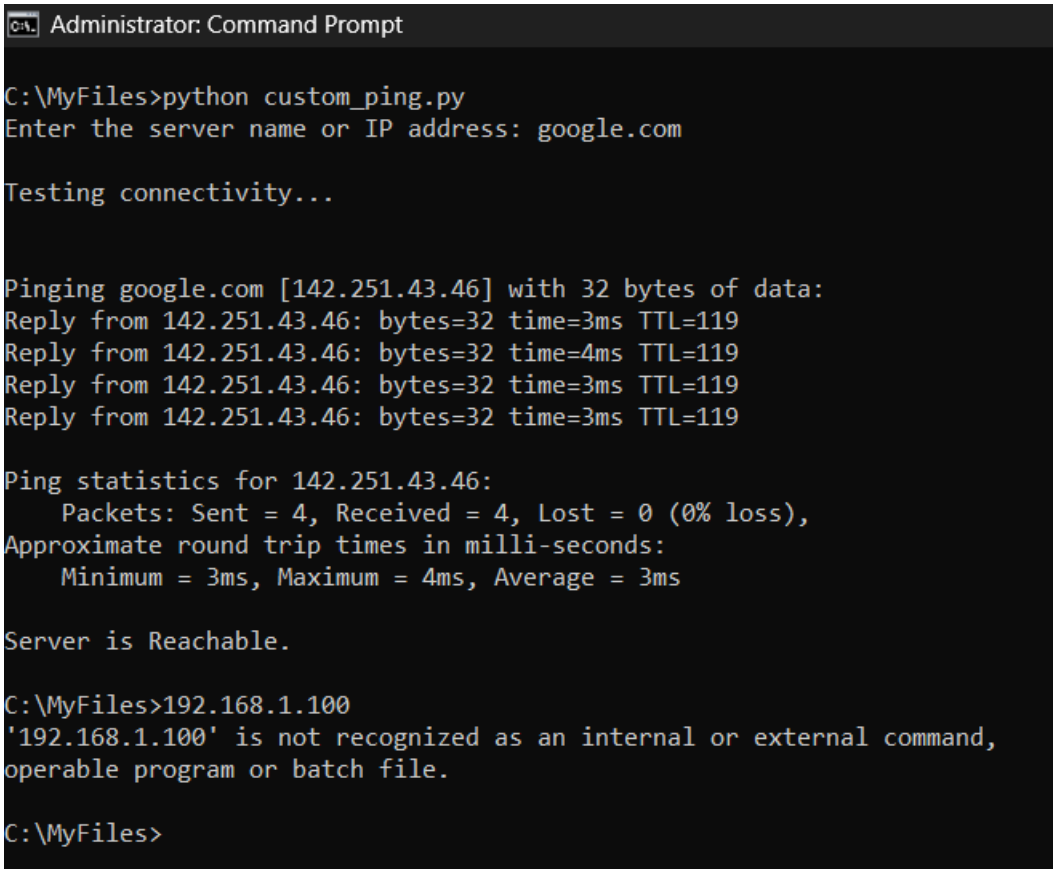
host = input("Enter the server name or IP address: ")
if platform.system().lower() == "windows":
    command = ["ping", "-n", "4", host]
else:
    command = ["ping", "-c", "4", host]

print("\nTesting connectivity...\n")

result = subprocess.run(
    command,
    capture_output=True,
    text=True
)

print(result.stdout)

if result.returncode == 0:
    print("Server is Reachable.")
else:
    print("Server is Not Reachable.")
```


Administrator: Command Prompt
C:\MyFiles>python custom_ping.py
Enter the server name or IP address: google.com
Testing connectivity...
Pinging google.com [142.251.43.46] with 32 bytes of data:
Reply from 142.251.43.46: bytes=32 time=3ms TTL=119
Reply from 142.251.43.46: bytes=32 time=4ms TTL=119
Reply from 142.251.43.46: bytes=32 time=3ms TTL=119
Reply from 142.251.43.46: bytes=32 time=3ms TTL=119
Ping statistics for 142.251.43.46:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 3ms, Maximum = 4ms, Average = 3ms
Server is Reachable.
C:\MyFiles>192.168.1.100
'192.168.1.100' is not recognized as an internal or external command,
operable program or batch file.
C:\MyFiles>